



DSN AI Bootcamp 2026 Project Participation – Data Analytics Track

Answer & Submission Sheet

Participant Name	Mercy Ijegbai
Dashboard tool used	Microsoft Excel

Part A - Data Cleaning Review

Identify at least three meaningful data-quality issues you found. For each issue, explain what you found, what you did, and why your action was appropriate.

Data-quality issue	What you found	What you did	Why this action was appropriate
Exact duplicate records	I found 12 exact duplicate rows. The duplicated transactions had the same values across all 16 columns.	I made a copy of the raw sheet and removed the duplicate rows from the copy, leaving 420 unique records. I kept the original raw sheet unchanged.	The repeated rows were exact duplicates, so removing the extra copies prevents the same transaction from being counted twice while keeping the raw data available for verification.
Missing Cost_Per_Unit_NGN values	There were 31 missing cost values. 20 of the 81 Abuja transactions had missing costs.	I checked the same products in other records and filled the missing costs only where the product had a consistent recorded cost. For example, Standing Fan was consistently ₦39,000.	Using a verified product-level cost is more appropriate than treating a blank cost as zero, because zero would overstate profit.
Revenue inconsistencies	22 of 432 transactions (5.1%) had Reported Revenue values that did not match Units × Unit Price × (1 – Discount).	I created a Corrected Revenue field using the formula Units × Unit Price × (1 – Discount). I used this corrected field for the final analysis.	Revenue should be based on the transaction details when the reported value does not agree with the underlying calculation. This makes the final revenue and profit analysis more reliable.
Inconsistent category/state entries	I found inconsistent labels, including Lagos appeared as "Lagos", "Lagos " (trailing space) and "LAGOS"; Accessories appeared as "Accessories ", "accessories" and "Accessory". and a 'Lagoa' state entry.	I checked the affected records and corrected the labels in the cleaned copy so that the same category/state is grouped consistently.	Standardizing labels prevents the same category or market from being split into separate groups in PivotTables and helps the dashboard show the correct totals.

Which data-quality issue had the greatest impact on your analysis?

The 12 exact duplicate records had the greatest impact on the analysis because they were counted as extra transactions and increased the reported totals.

Explain how correcting that issue changed or affected the analysis.

After removing the duplicates and completing the other corrections, the final corrected profit was ₦14,590,425. The raw reported-profit figure was ₦16,617,270, so the final corrected profit was ₦2,026,845 lower. Duplicate records alone accounted for ₦1,415,000 of profit that was being counted twice, while filling the missing costs reduced profit by ₦921,600; the corrected revenue calculation added ₦309,755 back.

Part B - Dashboard Audit

For each management claim, state your verdict, provide the corrected result where applicable, and explain the evidence supporting your conclusion.

1. Abuja is the most profitable market.



Verdict

Incorrect

Corrected answer / result

Lagos was the most profitable market, with corrected profit of ₦4,579,900.

Explanation / evidence

After cleaning and calculating profit from corrected revenue and validated costs, Lagos had the highest profit. Abuja had ₦1,698,100, while Lagos had ₦4,579,900.

2. Electronics is the strongest-performing product category and should receive more budget.

Verdict

Misleading

Corrected answer / result

Electronics had the highest category profit at ₦8,986,500, but it also had the highest return rate at about 11.9%.

Explanation / evidence

The first part of the claim is supported because Electronics generated the most profit. However, saying it should receive more budget without considering its return rate is misleading. The return rate was about 11.9% for Electronics, compared with about 4.6% for Accessories and 7.5% for Home.

3. Higher discounting is improving profitability.

Verdict

Incorrect

Corrected answer / result

Profit margin decreased as the discount level increased. It fell from about 41.7% at 0% discount to about 16.3% at 25% discount.

Explanation / evidence

The corrected analysis shows the opposite pattern to the claim. Higher discount levels were associated with lower profit margins, so increasing discounts cannot be described as improving profitability from this dataset.

4. The Online channel has the lowest return rate.

Verdict

Incorrect

Corrected answer / result

In-store had the lowest return rate at 6.44%. Online was 9.70%, while WhatsApp was 11.90%.

Explanation / evidence

Online and In-store both recorded 13 returned transactions, but In-store had more total transactions (202 compared with 134). Therefore, its return rate was lower.

5. August is the strongest sales month.

Verdict

Incorrect

Corrected answer / result

May was the strongest sales month, with corrected revenue of ₦7,008,500. August recorded ₦3,167,425.

The monthly corrected-revenue analysis shows May as the highest month, not August. August was the lowest month in the corrected monthly trend.

6. Lagos is underperforming compared with other markets.

Verdict



Incorrect

Corrected answer / result

Lagos had the highest corrected profit at ₦4,579,900

The corrected state analysis does not support the claim that Lagos was underperforming. Lagos generated the highest total profit among the markets in the cleaned dataset.

Explanation / evidence

Lagos: revenue ₦13.69m (29% of total), profit ₦4.58m (31% of total), margin 33.4% (highest of all states), return rate 5.6% (second lowest), average discount 6.4%. The "underperforming" conclusion was an artefact of the label inconsistency ("Lagos", "Lagos ", "LAGOS"), which split Lagos into three small groups.

7. The reported Revenue field can be used directly for decision-making.

Verdict

Misleading

Corrected answer / result

The Reported Revenue field should not be used directly without validation. I used Corrected Revenue for the final dashboard.

Explanation / evidence

22 of 432 transactions (5.1%) had reported revenue that did not match $\text{Units} \times \text{Unit Price} \times (1 - \text{Discount})$. Cleaned reported revenue was ₦46,369,970, while corrected revenue was ₦46,679,725, a net difference of ₦309,755.

8. The dashboard is sufficiently reliable for management to act on without further data cleaning.

Verdict

Incorrect

Corrected answer / result

The dashboard was not reliable enough to act on without further cleaning and validation.

Explanation / evidence

I found 12 duplicate rows, 31 missing cost values, 22 revenue inconsistencies, and inconsistent category/state entries. These issues could affect revenue, profit and the conclusions drawn from the dashboard, so the data needed to be cleaned and validated first.

Part C - Corrected Dashboard

Insert a clear snapshot of your corrected dashboard below. The dashboard may be created in Power BI or Excel.

MANAGEMENT DASHBOARD

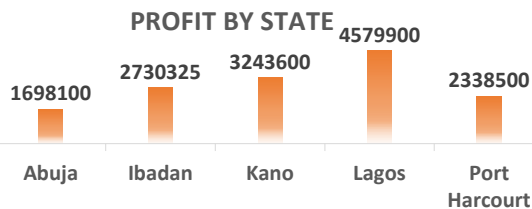
TOTAL REVENUE
₦46.68m

PROFIT
₦14.59m

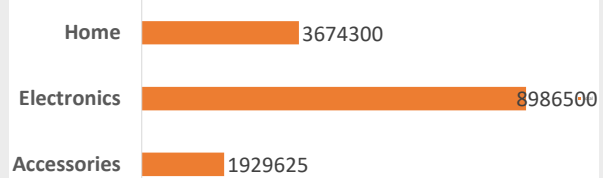
PROFIT MARGIN
31.3%

RETURN RATE
8.6%

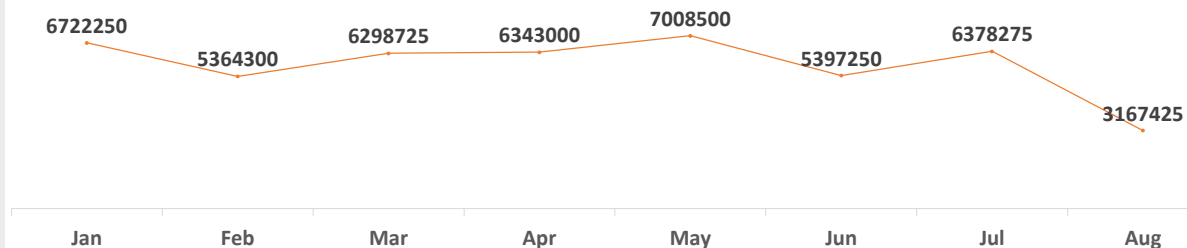
PROFIT BY STATE



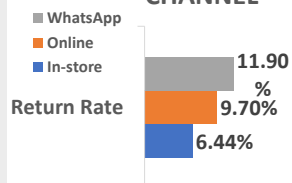
PROFIT BY CATEGORY



MONTHLY REVENUE TREND



RETURN RATE BY CHANNEL



KEY INSIGHTS

- 01 Lagos generated the highest profit.
- 02 Electronics generated the highest category profit, but also had the highest return rate.
- 03 Higher discounts were associated with lower profit margins.
- 04 Abuja had the lowest average profit margin (25%) while In-store had the lowest return rate (6.4%)

PROFIT MARGIN PER DISCOUNT LEVEL





Part D - Final Analysis Questions

1. Which conclusion in the original dashboard was the most misleading, and why?

Answer

The claim that the Reported Revenue field could be used directly for decision-making was the most misleading.

Explanation / evidence

It looked like a normal revenue field, but 22 of the 432 transactions (5.1%) had values that did not agree with the underlying Units, Unit Price and Discount. Using it without validation could affect profit calculations and the decisions based on them.

2. What changed most materially after you cleaned and corrected the data?

Answer

The most material change was in the final profit figure.

Explanation / evidence

The raw reported-profit total was ₦16,617,270, while the cleaned and corrected profit was ₦14,590,425, a decrease of ₦2,026,845. Duplicate removal, missing-cost corrections and revenue corrections all contributed to this change.

3. If management had acted on the original dashboard, what decision could have created the greatest business risk?

Answer

Increasing the budget for Electronics without first looking into its return rate could create the greatest business risk.

Explanation / evidence

Electronics generated the highest category profit at ₦8,986,500, but it also had the highest return rate at about 11.9%. Increasing spending based only on its profit could overlook the cost and business impact of those returns.

4. Based on your corrected analysis, what is the most important driver of business performance in this dataset?

Answer

Discount level appears to be an important driver of profitability in this dataset.

Explanation / evidence

The corrected analysis showed profit margins falling as discounts increased, from about 41.7% at 0% discount to about 16.3% at 25% discount. This shows a clear relationship in the dataset, although it does not by itself prove that discounting causes the change.

5. What three actions should management take next based on the evidence from your corrected dashboard?

Answer

1. Review and tighten the discount strategy, especially at the higher discount levels.
2. Investigate product returns, particularly in Electronics and across the channels with higher return rates.
3. Put a simple data-validation process in place for duplicates, missing costs, revenue calculations and inconsistent labels before future dashboards are used.

Explanation / evidence

These actions are based on the corrected dashboard: higher discounts were linked with lower margins, Electronics generated the most profit but also had the highest category return rate, and the audit found several data-quality issues that affected the original analysis.

Submission reminder: Complete all sections, insert a readable corrected-dashboard snapshot, ensure your written figures match the dashboard, and keep your Power BI or Excel working file available if verification is requested.